CHAPTER 3 · CONCEPT 1

Early Number Systems

Before our modern digits, people counted with tally marks, Roman numerals and Egyptian symbols. Learn how each works and how to convert to and from the numbers we use today.

Three early systems

1. Tally marks

One mark for each object — the oldest method. Usually grouped in fives (four lines and a cross stroke) to make them easy to read. Simple, but very long for big numbers.

2. Roman numerals

Seven letters stand for landmark numbers:

I	V	X	L	C	D	M
1	5	10	50	100	500	1000

- **Add** when a smaller (or equal) value follows a larger one: VI = 5 + 1 = 6, XXVII = 27.
- **Subtract** when a smaller value comes *before* a larger one — only in these six pairs: **IV = 4, IX = 9, XL = 40, XC = 90, CD = 400, CM = 900**.

2367 = MM (2000) + CCC (300) + LX (60) + VII (7) = MMCCCLXVII

3. The Egyptian system (base 10)

Special symbols for 1, 10, 100, 1000, … — each landmark number is 10 times the previous one (so all are powers of 10). A number is written by repeating these symbols and adding them up. For example, 324 = three "100"s + two "10"s + four "1"s.

Big idea: Roman and Egyptian numerals are **additive** — you read a number by adding up the symbol values. There is no "place value": the symbol means the same wherever it appears.

Prerequisite refresher: A "landmark number" is a value that gets its own symbol (like 1, 5, 10, 50…). "Additive" means you add the symbols' values to find the number.

Where marks slip away

Trap 1 — Only six subtractive pairs. IV, IX, XL, XC, CD, CM. There is no "IL" for 49 — it's XLIX (40 + 9).

Trap 2 — Don't repeat four times. A symbol repeats at most three times. 4 is IV, not IIII; 40 is XL, not XXXX.

Trap 3 — Work from the largest landmark down. Take as many 1000s as possible, then 500s, then 100s, and so on.

Trap 4 — Additive systems have no zero and no place value. The same symbol has the same value anywhere it appears.

Slip 5 — Read subtractive pairs as one unit (e.g. CM = 900) before adding the rest.

Model answers — write them like this

Example A. Write 27 in Roman numerals.

Step 1: Group from the largest landmark: 27 = 10 + 10 + 5 + 1 + 1.

Step 2: Replace with symbols: X X V I I.

∴ 27 = XXVII

Example B. Write 2999 in Roman numerals.

Step 1: 2999 = 2000 + 900 + 90 + 9.

Step 2: 2000 = MM, 900 = CM, 90 = XC, 9 = IX.

∴ 2999 = MMCMXCIX

Example C. What number is DCCXV?

Step 1: Read the symbols: D = 500, C = 100, C = 100, X = 10, V = 5.

Step 2: Add them: 500 + 100 + 100 + 10 + 5 = 715.

∴ DCCXV = 715

Example D. How is 324 written in the Egyptian (base-10) system?

Step 1: Break into powers of 10: 324 = 100 + 100 + 100 + 10 + 10 + 4.

Step 2: Use three "hundred" symbols, two "ten" symbols, and four "one" symbols.

∴ three 100s + two 10s + four 1s

Next: try the worksheet, then check the answer key.